Dae Seok Song

Seoul, Republic of Korea

AI Engineer | Medical AI and Modular Neural Networks

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Research Profile

AI engineer who has led model development, compression, clinical-product integration, and international regulatory evidence generation at a medical-AI startup since November 2022. My research asks whether a functional module keeps its role when moved into a separately trained model (so far tested only across models that differ only in initialization), and tests this with intervention: swapping a module between separately trained models and measuring what breaks (preliminary; removal is used only as a control in a separate setup).

Research Interests

Functional-module transplant and intervention; representation alignment; modular and continual learning; model compression and structured pruning; local learning rules.

Independent Research

RevolverNet: Detachable Functional Modules and Conditional Routing 2026–Present

Independent Researcher Seoul, Korea

- Designed a modular architecture with interchangeable slots, detachable functional modules, and a router.
- Across two completed image-classification conditions, one-time representation alignment recovered 99.4–99.8% of the performance lost under naive swaps, covering every directed donor-to-recipient pair among four separately trained models (12 ordered pairs per condition, below my pre-set 20-pair minimum). The models shared the dataset, architecture, and role assignment (roles were assigned by me, not learned) but differed in initialization. Preliminary, unpublished independent work; prior work on model stitching and re-basin already aligns independently trained networks, so the recovery itself is not claimed as new.
- In a separate setup that loads multiple modules together without the router (not the four transplant models above), measured the residual task-performance loss after removing a single module across designs matched on accuracy (0.03–0.77; preliminary, single seed, noise floor $\pm$0.019). I use removal only as a control, not as standalone evidence that a module is load-bearing.
- Implemented routing so that a router selects one module for each input.
- Long-term goal: let people inspect, remove, and replace functional modules while preserving validated behavior.

Brain-Inspired vs. Backpropagation Learning-Rule Comparison 2026

Independent Researcher Seoul, Korea

- Compared a burst-plasticity-style local learning rule against backpropagation on the same modular architecture, implemented at the firing-rate level (did not reach a spiking implementation).
- Backpropagation led on both accuracy (three-seed average 90.8% vs. 77.5%) and post-transplant module performance; the result redirected the project from "which rule learns better" toward intervention-based measurement of what a learned module actually preserves.

Professional Experience

Connecteve Co., Ltd. Nov 2022–Present

AI Engineer / Early Team Member Seoul, Korea

- Restructured an 11-model knee X-ray pipeline into 3 models (two uncertainty-aware, partial-label multi-task models plus one retained keypoint model): stored weights reduced by about 11$\times$ and wall-clock inference time per image 2.5$\times$ lower, with the overall F1 difference under 0.004 versus the baseline model (single seed).
- Rebuilt the optimization-to-serving line (ONNX optimization, 16-bit inference, BentoML serving): engine size 47% smaller and inference 3$\times$ faster, with clinical validation showing only a 0.06-percentage-point accuracy drop and 2.1% grade changes (non-inferiority confirmed by patient-level bootstrap).
- Tuned per-grade decision thresholds with Youden's J statistic across four tie-break strategies, surpassing an FDA-cleared competitor product's F1 on a matched evaluation.
- Built the standard PyTorch Lightning + Hydra experiment pipeline for medical foundation models (DINOv3, Rad-DINO) with selectable linear probing or LoRA.

- Built an in-house MLOps console (GUI/CLI) with a two-stage hyperparameter search (LLM-agent proposals, then Optuna/TPE on plateau) that raised detection mAP from 0.80 to 0.87; evaluated 12 agentic-workflow frameworks and selected LangGraph, running agents on vLLM (Qwen, Mistral); caching cut re-onboarding from 3 h 36 min to 38 min (5.7×).
- Led AI technical documentation for medical-device approval in Korea (MFDS), the EU (CE MDR), the UAE (DOH), and Singapore (HSA).

Minerva Soft Mar 2022–Jul 2022
Image Processing Intern Korea

- Replaced full-image barcode scanning with an OpenCV ROI pipeline, raising recognition from 70% to 90%; shipped a C++ DLL after 170,000+ memory-leak and stability iterations.

Education

University of Suwon Mar 2017–Feb 2023
B.S. in Information Security, Valedictorian GPA 4.29/4.50; Major 4.37/4.50

Selected Publications

Lee, D. W., **Song, D. S.**, et al. "Accurate, automated classification of radiographic knee osteoarthritis severity using a novel method of deep learning: Plug-in modules." *Knee Surgery & Related Research*, 2024. Second author; devised and implemented the core plug-in-module ensemble that imitates clinicians' consensus, with the paper reporting an accuracy improvement of about 6%. doi:10.1186/s43019-024-00228-3

"Image size transformation data augmentation method for lesion segmentation on normalization and augmentation medical images." *Korea Software Congress (KSC)*, 2022, poster presentation. First author.

Patents & Other Research Output

- Inventor, Korean Patent Application 10-2024-0124529 and PCT/KR2025/012689: ensemble system for knee-osteoarthritis severity classification (the plug-in-module method above).
- Inventor, Korean Patent Application 10-2025-0043898: generating a patient's projected knee image from the current radiograph and its K–L/OARSI grades.

Technical Strengths

Research	Controlled ablation design, multi-seed evaluation, intervention-based functional analysis, representation analysis, patient-level bootstrap non-inferiority tests, reproducibility
ML	PyTorch, PyTorch Lightning, Hydra, multi-task and partial-label learning, foundation-model adaptation (linear probing, LoRA), medical-image classification/segmentation
Compression & serving	ONNX, OpenVINO 16-bit, quantization and structured/block pruning, TorchServe, BentoML, FastAPI, vLLM
Engineering	Python, C/C++, Docker, GPU server operation (H100, L40S), regulated-product documentation

Languages

Korean (native); English (conversational); Japanese (conversational).